

From: "Shapiro2,E" <E.Shapiro2@lse.ac.uk>
Subject: Re: European Innovation Council Application.
Date: 20 February 2026 at 11:25:33 GMT
To: Francesco Caselli <F.Caselli@lse.ac.uk>, Ehud Shapiro
<udi.shapiro@gmail.com>

Hi Francesco,

Your questions are profound and deep and require discussion, which I will be happy to have:)

Briefly, my view on them:

- Any technology can be used and abused. Recent examples are blockchain and AI. Blockchain/Bitcoin were created with the best of intents (truly), but it is the preferred and most effective way today for money laundering and trafficking, serving organised crime around the world.
- AI is extremely dangerous, can be used by bad actors to create all sorts of terrible weapons and attacks on humanity.

Regarding grassroots technology, indeed it can, but for organising criminal activities there are already good alternatives today (e.g. Signal).

What grassroots could do (and for me it is an intended consequence, not an accident) is to serve people to communicate, organise, and maybe even have barter commerce under repressive regimes that control the Internet in their country. Specifically, if grassroots social networking was already deployed, the fate of the citizens' rebellion in Iran might have been different, given how the Internet blackout by the government helped quell it.

Regarding programming languages, they are never "essential", by definition, since they are compiled into machine language. So, in the end, they do not "exist" in running software.

They are always a convenience, serving various productivity and correctness and security goals. I started working on GLP after seeing how the MSc student at the Technion was struggling to implement a grassroots social graph in a "regular" programming language (Python), and how ugly and unwieldy the result was.

More specifically, grassroots apps present a tremendous programming challenge (being peer-to-peer and not client-server). My belief (which still has to be corroborated, but initial evidence is promising) is that AI will find it much easier and productive to program them in a high-level mathematically- and logically-oriented programming language, designed specifically for their implementation. Such languages have been around for a while (I invented an influential one in 1982; but it is too "powerful" for our needs; GLP is a simplified and cleaner cousin), but did not catch on, partly because the "average" programmer found

them hard to use. But AI is not your average programmer. Is it your best programmer. And it becomes better by the day.

Will be happy to continue this...

Udi

From: Francesco Caselli <F.Caselli@lse.ac.uk>
Sent: 20 February 2026 10:55
To: Shapiro2,E <E.Shapiro2@lse.ac.uk>; Ehud Shapiro
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Actually that reminds me of another thing I am not clear about.

I understand the benefit of a technology that allows a network of phones to connect without relying on third party infrastructure or verification. But why does this require a new programming language?

From: Shapiro2,E <E.Shapiro2@lse.ac.uk>
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BTW, i (and Claude) are developing the grassroots language GLP using Dart, developed by Google.